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Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization

Submitted As

Final Project Report

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To

BSSE 4th Year Exam Committee
Institute of Information Technology
University of Dhaka

Subject: Submission of Final Project Report - *APEX-Net: Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization*

Sir,

I am respectfully submitting the final project report for my Software Project (SE 801) titled “APEX-Net: Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization.”

This report comprehensively covers the project's requirement specifications, technical details, including design, implementation, and testing phases.

I have made every effort to ensure the quality and completeness of this work. However, I will be grateful for any valuable feedback or suggestions that may help improve the project further.

Thank you for your time and kind consideration.

Sincerely,

Habibur Rahman Mahin

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Supervisor's Signature

Acknowledgement

I am grateful to the Institute of Information Technology, University of Dhaka, for providing me with the opportunity to work on the project “APEX-Net: Anatomically Prior-Enhanced X-Ray Network for Thoracic Disease Classification and Localization.” I would like to express my sincere gratitude to my supervisor, Mohd. Zulfiquar Hafiz, Professor, Institute of Information Technology, University of Dhaka, for his continuous guidance, valuable suggestions, and support throughout the development of this project. Their insights and feedback played an important role in shaping the direction and quality of this work. Lastly, I would like to thank my classmates for their cooperation, discussions, and helpful inputs during different phases of the project.

Abstract

APEX-Net is a web-based intelligent diagnostic system designed to automate the screening of chest X-ray images across fourteen thoracic diseases. The system integrates a deep learning model based on DenseNet121 enhanced with an Anatomical Prior Attention Module (APAM) to produce multi-label predictions alongside visual heatmaps that highlight regions of concern. The platform is developed to be accessible to general users, patients, and medical trainees, particularly in low-resource environments where expert radiological support is limited.

The system combines modern web technologies with advanced artificial intelligence to deliver a scalable, interpretable, and efficient diagnostic tool. This report presents the system's design, architecture, implementation, and evaluation in a structured and comprehensive manner.

Table of Contents

Acknowledgement.....	3
Abstract.....	4
Table of Contents.....	5
1. Introduction.....	7
1.1 Background.....	7
1.3 Objectives.....	8
1.4 Scope of the System.....	8
1.5 Timeline.....	8
2. Literature Review.....	9
3. System Overview.....	9
3.1 General Description.....	9
3.2 System Workflow.....	9
3.3 Disease Categories.....	10
3.4 Output Interpretation.....	10
4. Elicitation of APEX-Net.....	10
4.1 Collaborative Requirements Gathering.....	10
4.2 Viewpoints of Stakeholders.....	11
4.3 QFD(Quality Function Deployment).....	12
4.4 User Story.....	14
5. Database Modeling.....	15
5.1 Schemas of APEX-Net.....	15
5.2 ER Diagram.....	17
6. Architectural Design.....	18
6.1 Architectural Context Diagram.....	18
6.2 Defining Archetypes.....	19
7. Component Level Design.....	23
7.1 Top Level Components.....	23
7.2 Mapping Requirements to Software Architecture.....	25
8. Deployment.....	27
9. Applied Methodology.....	29
9.1 Workflow.....	29
9.2 Input.....	30
9.3 Processing.....	31
9.4 Output.....	34
10. User Manual and User Interface.....	38
10.1 Introduction.....	38
10.2 System Requirements.....	38

10.3 Accessing APEX-Net.....	38
10.4 Registering a New Account.....	39
10.5 Logging In.....	40
10.6 The Application Layout.....	41
10.7 Uploading and Analyzing an X-ray.....	42
10.8 Understanding the Results.....	43
10.9 Scan History.....	45
10.10 Updating Your Profile.....	46
10.11 Logging Out.....	47
10.12 Common Messages and Troubleshooting.....	47
11. Preliminary Test Plan.....	48
11.1 High-Level Description of Testing Goals.....	48
11.2 Item Pass/Fail Criteria.....	48
12. Conclusion.....	50
13. References.....	51

1. Introduction

1.1 Background

Thoracic diseases such as pneumonia, pneumothorax, and cardiomegaly remain among the leading causes of mortality worldwide. Chest X-ray imaging is one of the most widely used diagnostic tools for detecting such conditions due to its affordability and accessibility. However, accurate interpretation of X-rays requires trained radiologists, whose availability is limited in many regions.

Recent advances in deep learning have enabled automated image analysis systems capable of assisting in medical diagnosis. Despite these advancements, existing systems often lack interpretability, struggle with rare disease detection, and fail to incorporate anatomical context.

1.2 Problem Statement

As identified in the project proposal, current AI-based chest X-ray systems suffer from three major limitations:

1. **Imbalanced Detection**

Models tend to perform well on common diseases but fail to detect rare but critical conditions.

2. **Lack of Anatomical Awareness**

Most models treat the image uniformly without focusing on medically relevant regions.

3. **Lack of Interpretability**

Predictions are often presented without visual explanations, reducing clinical trust.

APEX-Net addresses these issues by incorporating anatomical priors, improving rare disease detection, and providing heatmap-based explanations.

1.3 Objectives

The primary objectives of APEX-Net are:

- 1. To develop an anatomically-aware deep learning model for chest X-ray analysis
- 2. To ensure reliable detection across both common and rare thoracic diseases
- 3. To provide interpretable outputs using heatmaps
- 4. To design a scalable and user-friendly web application
- 5. To enable deployment in low-resource environments

1.4 Scope of the System

The system is intended for:

- General public health awareness
- Medical students and trainees
- Preliminary screening assistance

It is important to note that APEX-Net is **not a replacement for professional medical diagnosis**, but rather a supportive tool.

1.5 Timeline

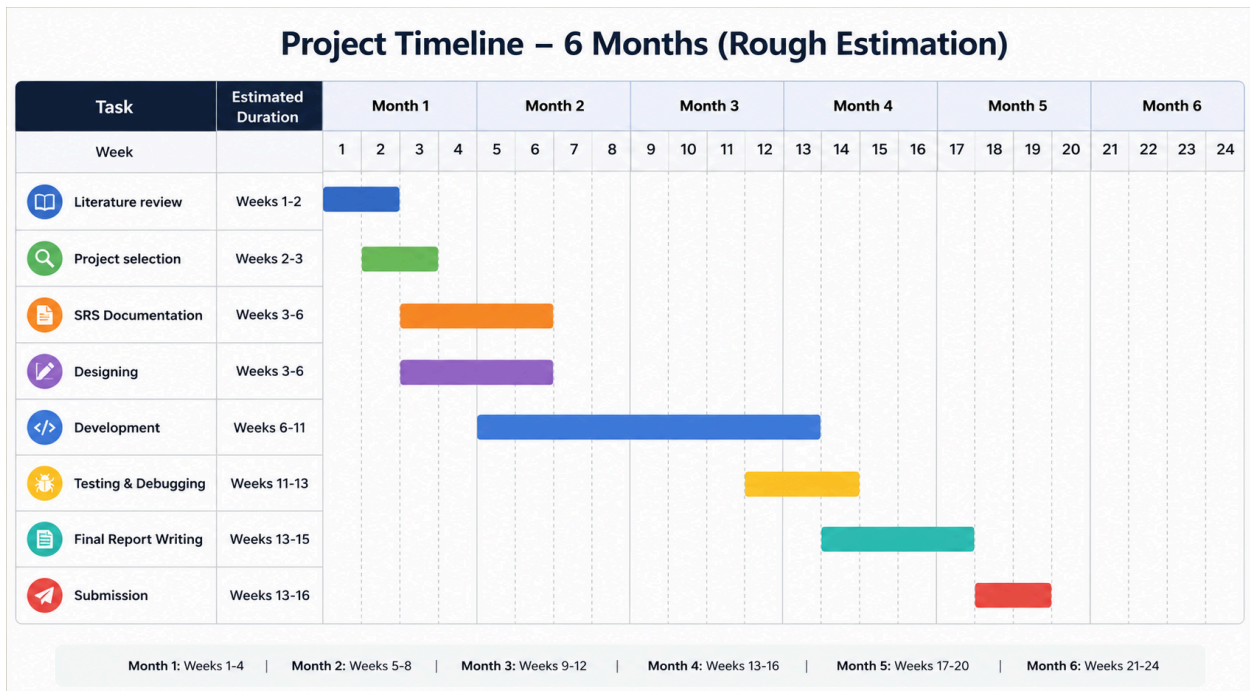


Figure-1 : Timeline of Project

2. Literature Review

Recent research in chest X-ray classification has focused heavily on deep convolutional neural networks. DenseNet-based architectures have shown strong performance due to their ability to reuse features and maintain gradient flow.

However, studies have highlighted limitations such as:

- Poor generalisation to rare conditions
- Lack of explainability
- Absence of anatomical reasoning

Attention-based models, such as those incorporating anatomical prior maps, have shown promise in addressing these issues. APEX-Net builds upon these advancements by integrating an Anatomical Prior Attention Module (APAM) into a DenseNet121 backbone.

3. System Overview

3.1 General Description

APEX-Net is a web-based system that enables users to upload chest X-ray images and receive automated diagnostic results. The system processes the image through a trained deep learning model and returns predictions along with a heatmap.

3.2 System Workflow

1. User registers or logs into the system
2. User uploads a chest X-ray image
3. Backend validates and preprocesses the image
4. Model performs inference
5. Heatmap is generated
6. Results are stored and displayed

3.3 Disease Categories

The system detects the following fourteen thoracic conditions:

Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, Pneumothorax

3.4 Output Interpretation

- Multi-label probability scores are generated
- If one disease probability is significantly higher, it is flagged as primary risk
- Heatmaps visually highlight affected regions

4. Elicitation of APEX-Net

The main task of the elicitation phase is to combine the elements of problem solving, negotiation, and specification. A collaborative working approach among stakeholders is essential to effectively gather and refine system requirements. For APEX-Net, multiple discussions and iterative feedback sessions were conducted to understand the needs of different user groups and ensure that the system design aligns with real-world expectations.

The following tasks have been completed for eliciting the requirements of APEX-Net:

- Collaborative Requirements Gathering
- Viewpoints of Stakeholders
- Quality Function Deployment
- User Story

4.1 Collaborative Requirements Gathering

During the inception phase of the APEX-Net project, several meetings and discussions were conducted with potential stakeholders to gather requirements. These sessions helped identify user expectations, system constraints, and functional priorities. Through continuous interaction and feedback, requirements were refined and negotiated to ensure feasibility and usability.

The stakeholders of APEX-Net include:

1. General Users (Patients / Public Users)
2. Medical Students and Trainee Doctors
3. Healthcare Professionals and Researchers

4.2 Viewpoints of Stakeholders

General Users (Patients / Public Users)

General users interact with APEX-Net as a diagnostic assistance tool. They expect a simple and intuitive interface where they can upload chest X-ray images and receive quick and understandable results. These users prioritise ease of use, fast response time, and clarity in output presentation.

Since most users may not have a medical background, the system must present predictions in a simplified manner. The inclusion of visual heatmaps helps users understand which areas of the X-ray are of concern. Additionally, users expect strong privacy and security measures to protect their medical data. The system serves as an initial screening tool that raises awareness and encourages timely medical consultation when necessary.

Medical Students and Trainee Doctors

Medical students and trainee doctors use APEX-Net as both a learning and diagnostic support tool. They are interested in understanding how artificial intelligence models interpret chest X-rays and identify different thoracic conditions.

The system provides probability distributions for multiple diseases, allowing students to analyse diagnostic uncertainty and compare different conditions. The heatmap visualisations further enhance their learning by showing region-specific importance, helping them correlate theoretical knowledge with practical image interpretation.

Access to historical scan data also enables them to study patterns and improve their clinical reasoning skills. Thus, APEX-Net acts as an educational platform that complements traditional medical training.

Healthcare Professionals and Researchers

Healthcare professionals and researchers focus on the reliability, consistency, and interpretability of the system. They expect accurate predictions across both common and rare diseases, as well as transparency in how results are generated.

The heatmap-based visual explanations are particularly valuable, as they provide insight into the model's decision-making process. This improves trust and allows professionals to validate the AI's output against clinical expectations.

Researchers may also use the system to analyse performance trends, evaluate model behaviour, and explore improvements in AI-based diagnostic systems. APEX-Net therefore serves as a foundational tool for further research and innovation in medical imaging.

4.3 QFD(Quality Function Deployment)

Quality Function Deployment (QFD) is a technique that translates the needs of the customers into technical requirements for software. Ultimately the goal of QFD is to translate subjective quality criteria into objective ones that can be quantified and measured, and can then be used to design and manufacture the product. It is a methodology that concentrates on maximizing customer satisfaction from the software engineering process. So, we have followed this methodology to identify the requirements for the project. The requirements which are given below, are identified successfully by the QFD.

4.3.1 Normal Features

Chest X-ray Upload and Analysis:

Users can upload chest X-ray images (JPG, PNG, or DICOM) through a simple web interface. The system processes the image and performs automated analysis using the trained deep learning model.

Multi-label Disease Prediction:

The system predicts probabilities across 14 thoracic diseases simultaneously. It highlights the most dominant condition when one disease probability is significantly higher.

Heatmap Visualization (Explainability):

For each prediction, the system generates a heatmap overlay that highlights the regions of the X-ray influencing the diagnosis, improving interpretability and user understanding.

Scan History Management:

Users can view previously uploaded scans along with timestamps, predictions, and heatmaps. This allows tracking and comparison of results over time.

4.3.2 Expected Features

Registration and Authentication:

Users register and log in using email and password. Authentication ensures secure access to personal scan history and protects user data.

Image Preprocessing and Validation:

Uploaded images are automatically validated and preprocessed (resizing, normalization) before being passed to the model, ensuring consistent input quality.

Asynchronous Inference Processing:

For handling larger workloads or slower inference, the system supports asynchronous processing to maintain responsiveness and scalability.

4.3.3 Exciting Features

Anatomically-Aware AI Model (APAM Integration):

The model focuses on anatomically relevant regions of the chest, improving detection accuracy by mimicking how radiologists interpret X-rays.

Rare Disease Sensitivity Optimization:

The system is designed to detect both common and rare thoracic conditions effectively through balanced training and per-disease tuning.

User-Friendly Diagnostic Interface:

Results are presented in a clear and structured format, combining probability scores with visual explanations, making the system accessible even to non-expert users.

4.4 User Story

APEX-Net aims to provide an accessible and intelligent solution for early detection of thoracic diseases through AI-powered chest X-ray analysis. By combining automated diagnosis with visual explanations, the system helps both medical professionals and general users understand results more clearly and supports decision-making in low-resource environments.

Upload and AI-Based Diagnosis

Users can upload a chest X-ray image to the platform and receive instant AI-powered analysis across 14 thoracic conditions. The system processes the image and provides predictions quickly, making it useful as a first-pass screening tool, especially in areas with limited access to radiologists.

Visual Heatmap Explanation

For each diagnosis, the system generates a visual heatmap highlighting the regions of concern in the X-ray. This allows users—including non-medical users—to understand where abnormalities are detected, improving interpretability and trust in the results.

Smart Anatomical Focus

The AI model focuses on anatomically relevant regions of the chest instead of analysing the entire image uniformly. This mimics how radiologists interpret X-rays and improves diagnostic accuracy by prioritising important areas.

Rare Disease Detection

The system is designed to detect not only common thoracic conditions but also rare and serious diseases. A customised training approach ensures balanced performance across all conditions, addressing limitations of traditional models.

Clinical Reliability and Accuracy

APEX-Net applies advanced training techniques and per-disease sensitivity tuning to ensure consistent and clinically meaningful results across all 14 conditions, improving reliability in real-world usage.

Lightweight Web Application

The platform is delivered as a lightweight web application, making it accessible and deployable in low-resource clinical environments. Users can access the system easily without requiring specialised hardware or software.

5. Database Modeling

Data based modeling is a visual representation of a database structure and how data is organized within it. It involves creating entity-relationship diagrams (ERDs) that depict tables (entities), attributes (columns), and the relationships between them. Database modeling is typically achieved by analyzing the data requirements of a system, identifying entities and their attributes, and then using modeling tools or software to create ERDs that serve as a blueprint for database design and implementation.

5.1 Schemas of APEX-Net

Table-1: Schema attributes

Table	Attribute Name	Data Type
Profiles	id (PK)	UUID
	full_name	TEXT
	created_at	TIMESTAMPTZ
Xray_Uploads	id (PK)	UUID
	user_id (FK)	UUID
	file_name	TEXT
	file_size	INT4
	storage_path	TEXT
	uploaded_at	TIMESTAMPTZ

	created_at	TIMESTAMPTZ
Predictions	id (PK)	UUID
	user_id (FK)	UUID
	xray_upload_id (FK)	UUID
	prediction_data	JSONB
	confidence_score	FLOAT8
	analyzed_at	TIMESTAMPTZ
	created_at	TIMESTAMPTZ

5.2 ER Diagram

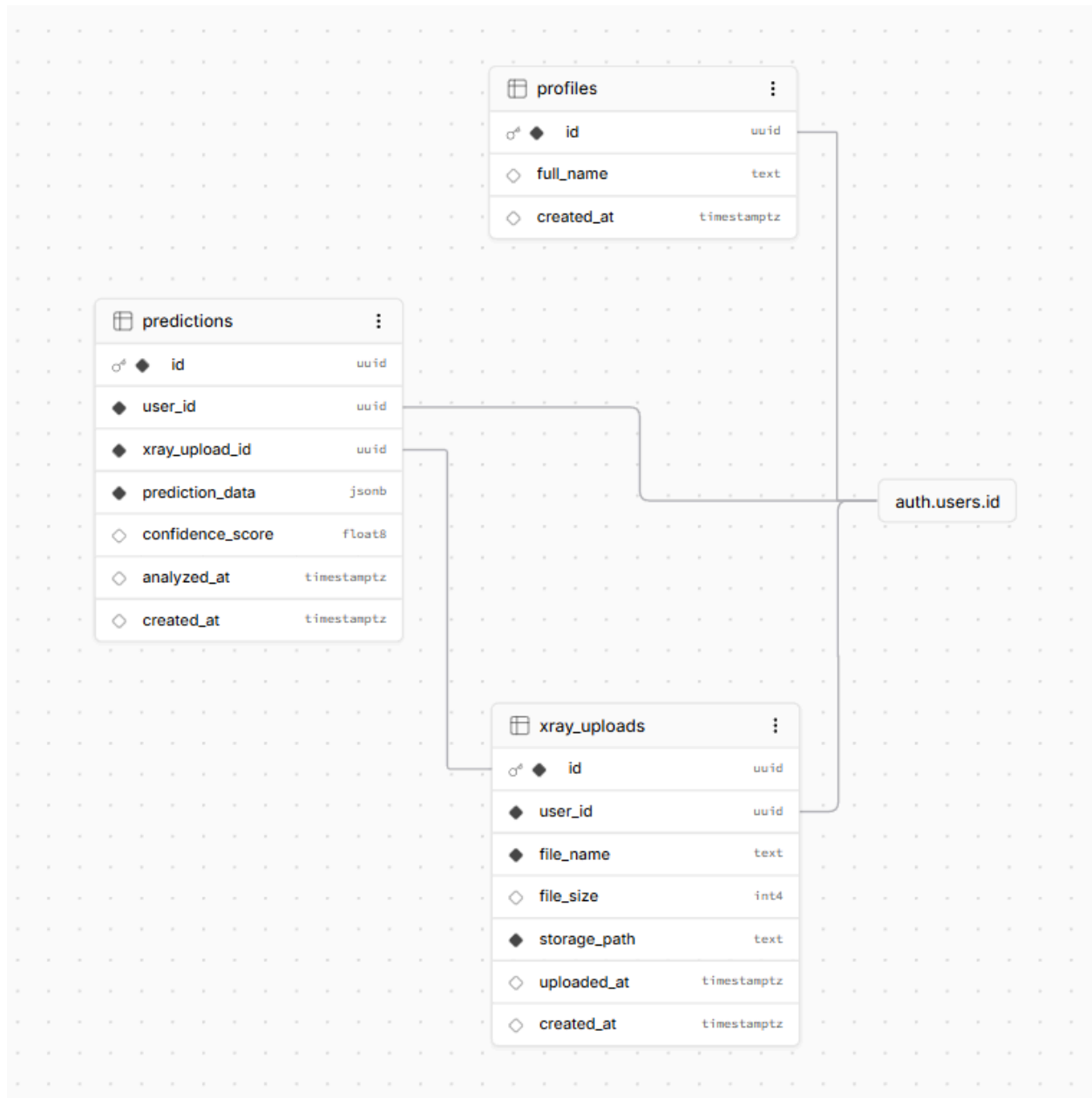


Figure-2 : ER Diagram of APEX-Net

6. Architectural Design

Architectural design represents the high-level structure of a software system and how its major components are organized. It identifies the key modules, defines their responsibilities, and describes how they interact with each other. For APEX-Net, the architecture must support a computationally heavy deep learning model, a responsive web interface, and secure storage of medical images and prediction results. A clear architecture also keeps the system scalable and maintainable, and provides a foundation for the detailed design and implementation decisions that follow.

6.1 Architectural Context Diagram

An Architectural Context Diagram illustrates how the target system interacts with external entities beyond its own boundary. External entities are grouped into the following categories:

- **Superordinate systems:** larger systems that use APEX-Net as a part of their own processing. APEX-Net is currently a stand-alone system and is not used by any superordinate system. Integration with hospital information systems is listed as future scope.
- **Subordinate systems:** systems on which APEX-Net depends to deliver its functionality. These are:
 - **Authentication and Email Service:** handles account creation, credential verification, session tokens, and the emails sent for account confirmation.
 - **PostgreSQL Database:** stores user profiles, X-ray upload records, and prediction results (the Profiles, Xray_Uploads, and Predictions tables described in Chapter 7).
 - **Object Storage:** stores the uploaded chest X-ray images and the generated heatmap overlays; the database keeps only the storage path of each file.
 - **AI Model:** the trained DenseNet121 model enhanced with the Anatomical Prior Attention Module (APAM), which produces the 14-class probability scores and the data used for heatmap generation.
- **Peer-level systems:** systems that exchange data with APEX-Net as equals. APEX-Net does not interact with any peer-level system.

- **Actors:** general users (patients and the public), medical students and trainee doctors, and healthcare professionals and researchers are the primary users of the system.

The architectural context diagram of APEX-Net is shown below.

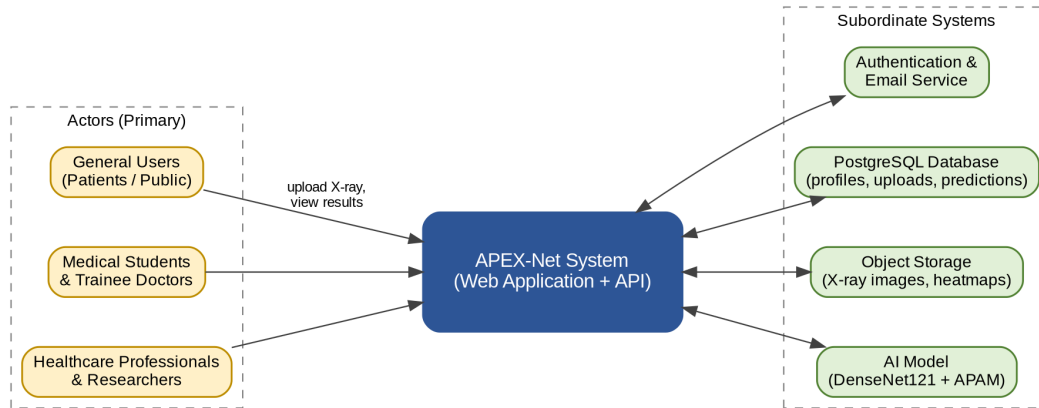


Figure-3 : Architectural Context Diagram

6.2 Defining Archetypes

APEX-Net combines two architectural patterns: a **Pipe and Filter** architecture that organises the image-analysis process, and a **Layered** architecture that organises the whole application. Together they form a pipeline that is easy to extend, test, and reason about.

6.2.1 Pipe and Filter Architecture

The core of APEX-Net is the process that turns an uploaded X-ray into a diagnostic result. This process requires sequential data transformation through several stages, which makes the Pipe and Filter pattern a natural fit. Each stage (filter) performs one specific operation and passes its output to the next stage through a connector (pipe). The stages are: validation of the uploaded file, pre-processing of the image, model inference, post-processing of the raw scores, heatmap generation, and persistence of the results.

Because every filter has a single responsibility and a well-defined input and output, a filter can be modified or replaced independently. For example, the model inference filter can be upgraded to a newer network without changing validation, heatmap generation, or storage. A failure in any filter, such as an unsupported file type in the first filter, stops the pipeline early and returns a clear error to the user.

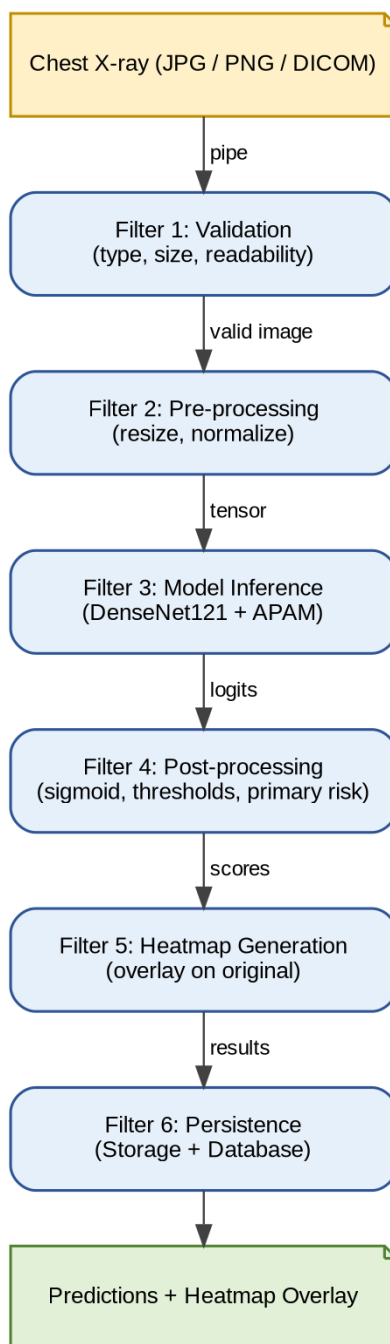


Figure-4 : Pipe and Filter Pattern for the X-ray analysis process

6.2.2 Layered Architecture Instantiation for APEX-Net

At the system level, APEX-Net follows a layered architecture in which each layer depends only on the layer directly below it. The four layers are described below.

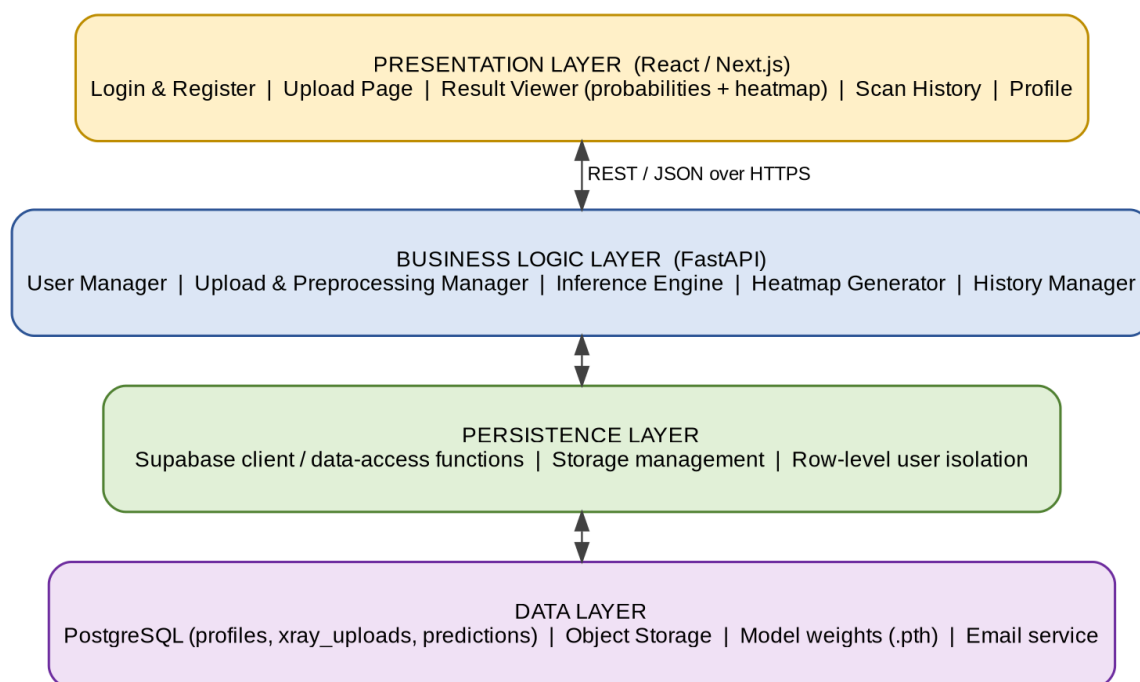


Figure-5 : Layered Architecture of APEX-Net

1. Data Layer

Components: PostgreSQL database, object storage, trained model weights, and the email service.

- Stores all user profiles, uploaded X-ray images, heatmap images, and prediction results.
- Holds the trained APEX-Net model weights that are loaded by the inference engine.
- Provides the email service used for account confirmation.

2. Persistence Layer

Components: data-access functions, storage management, and per-user access control.

- Provides create, read, and delete operations for uploads, predictions, and profile data.

- Maps domain objects (users, uploads, predictions) to database records and storage paths.
- Ensures that every query is restricted to the authenticated user, so one user can never read another user's scans.

3. Business Logic Layer

Components: User Manager, Profile Manager, Upload and Preprocessing Manager, Inference Engine, Heatmap Generator, and Scan History Manager.

- Validates and pre-processes uploaded images.
- Runs the APEX-Net model, converts raw outputs into probabilities, and identifies the primary risk.
- Generates heatmap overlays and coordinates saving of results.
- Handles slow inference asynchronously so that the interface remains responsive.

4. Presentation Layer

Components: login and registration pages, X-ray upload interface, result viewer, scan history page, and profile page.

- Provides a simple and intuitive interface for general users who may have no medical background.
- Displays the 14 disease probabilities together with the heatmap in a clear and structured format.
- Lets users browse previous scans with their timestamps, predictions, and heatmaps.

7. Component Level Design

7.1 Top Level Components

The top-level components of APEX-Net are illustrated below. They give a high-level overview of the structure and functionality of the system. The red arrows show the main analysis flow, from the uploaded image to the saved result.

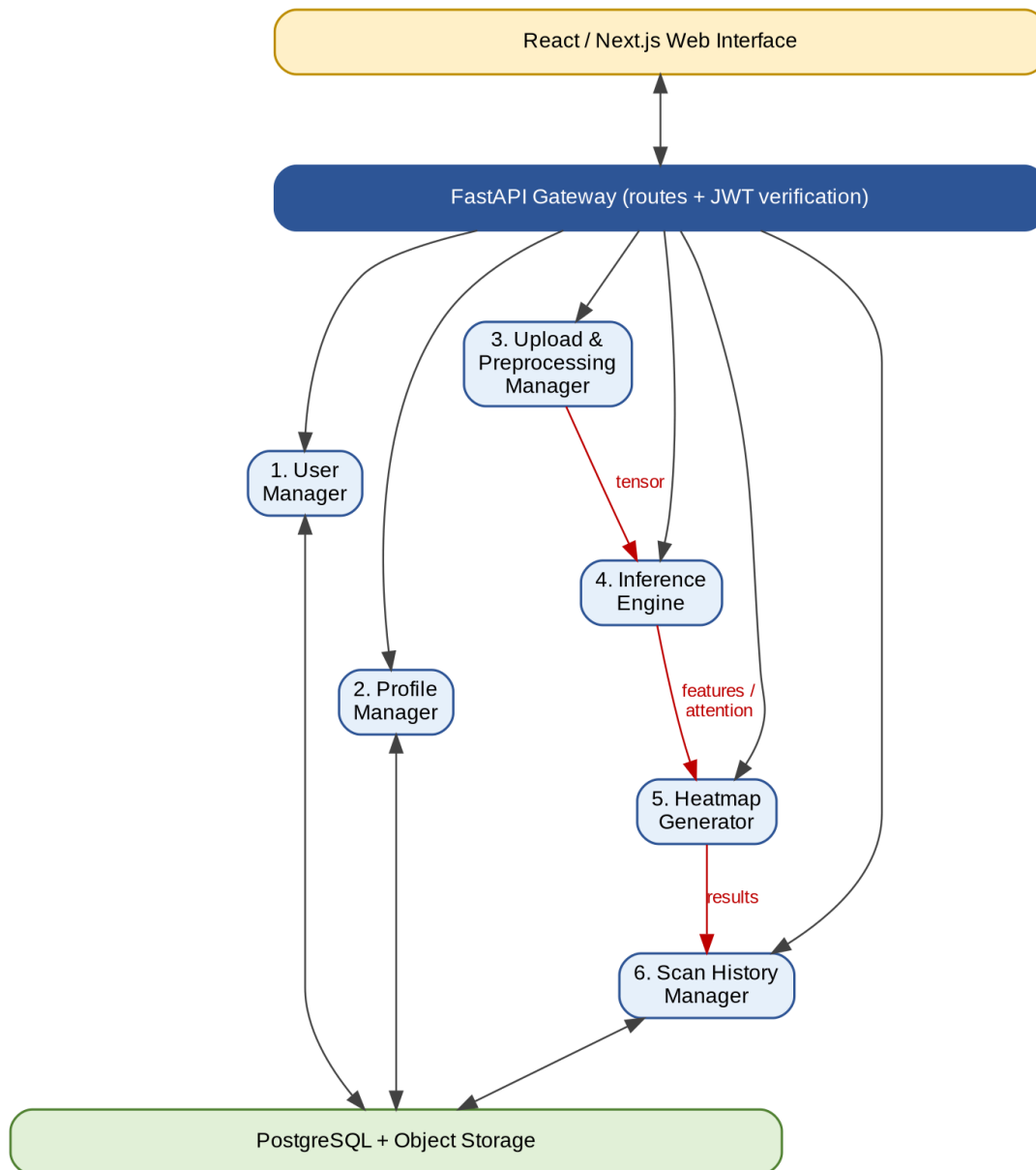


Figure-6 : Top Level Components

Component 1: User Manager

This component is the security gateway of the system. It handles user registration with email and password, sends the account confirmation email, verifies login credentials, and issues a signed session token. Every protected request carries this token, and the component rejects requests whose token is missing, invalid, or expired. It also supports secure sign-out.

Component 2: Profile Manager

This component manages the personal information of a user. It stores and displays the user's full name and allows the user to update the profile and change the account password. All profile data is linked to the authenticated user's identifier.

Component 3: Upload and Preprocessing Manager

This component receives the chest X-ray image from the user and validates its file type (JPG, PNG, or DICOM), size, and readability. Valid files are stored in object storage and an upload record is created. The image is then converted into the format expected by the model: it is resized, normalized, and turned into a tensor. Invalid files are rejected immediately with a descriptive error message.

Component 4: Inference Engine

This component loads the trained APEX-Net model (DenseNet121 backbone with the Anatomical Prior Attention Module) once at server start and runs it on each pre-processed image. It produces independent probability scores for the 14 thoracic diseases (multi-label classification), applies the per-disease decision thresholds, and flags the dominant condition as the primary risk when one probability is significantly higher than the others. Slow or concurrent requests are processed asynchronously.

Component 5: Heatmap Generator

This component produces the visual explanation of a prediction. Using the model's feature maps and attention, it computes which regions of the X-ray contributed most to a disease score, converts them into a colour heatmap, and overlays it on the original image. The overlay is saved to object storage so that it can be displayed again from the history page.

Component 6: Scan History Manager

This component provides persistent storage and retrieval of all analyses. It saves each upload record and its prediction (probabilities, confidence score, and timestamp) to the database, and lets the user list, open, and delete previous scans. Results are always filtered by the user's identifier to guarantee data isolation.

7.2 Mapping Requirements to Software Architecture

The following table shows how the requirements identified through QFD (Chapter 4) are matched to the architectural components.

Table-2: Matching requirements to software architecture

Requirement	Architectural Component(s)
Users can register with email and password and receive a confirmation email.	User Manager
Users can log in and receive a signed token for secure session management.	User Manager
Users can update their profile and change their password.	Profile Manager
Users can upload chest X-ray images in JPG, PNG, or DICOM format.	Upload and Preprocessing Manager
Unsupported or invalid files are rejected with a clear error message.	Upload and Preprocessing Manager
Uploaded images are automatically validated, resized, and normalized.	Upload and Preprocessing Manager
The system predicts probabilities for 14 thoracic diseases simultaneously.	Inference Engine
The dominant disease is highlighted as the primary risk.	Inference Engine
Rare diseases are detected through balanced training and per-disease tuning.	Inference Engine (APEX-Net model)

Requirement	Architectural Component(s)
The model focuses on anatomically relevant chest regions.	Inference Engine (APAM)
Slow or concurrent inference requests are handled asynchronously.	Inference Engine
A heatmap highlights the regions that influenced each prediction.	Heatmap Generator
Users can view previous scans with timestamps, predictions, and heatmaps.	Scan History Manager
Results and images are stored securely and are visible only to their owner.	Scan History Manager, User Manager

8. Deployment

A deployment diagram shows how the software components of APEX-Net are distributed over physical and virtual nodes at run time. APEX-Net is designed as a lightweight web application so that it can be deployed in low-resource environments: the user needs only a web browser, and all heavy computation runs on the server. The deployment diagram of APEX-Net is shown below.

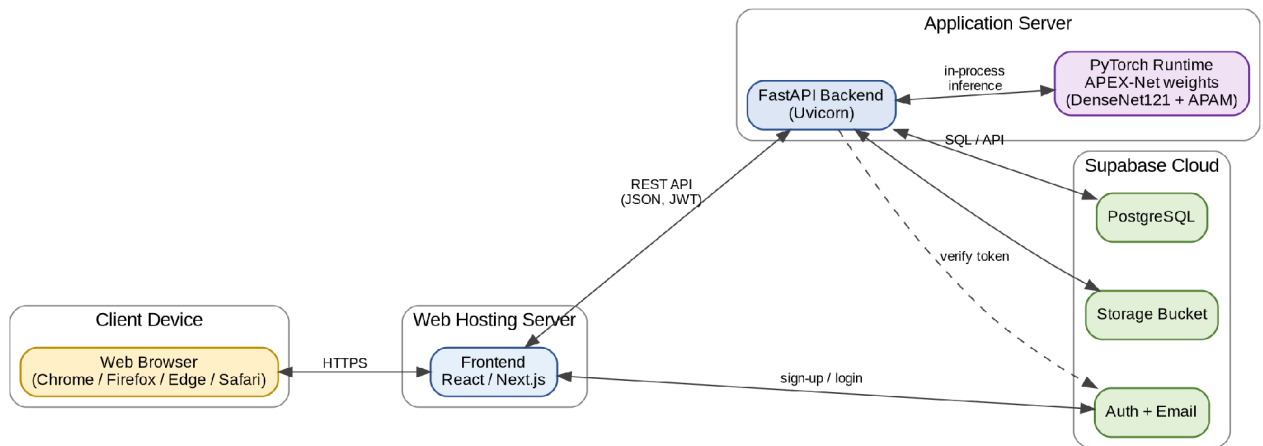


Figure-8 : Deployment Diagram of APEX-Net

The nodes of the deployment are described below.

Table-3: Deployment nodes

Node	Software Deployed	Responsibility
Client Device	Modern web browser (Chrome, Firefox, Edge, Safari)	Displays the interface, uploads images, and shows results. No installation is required.
Web Hosting Server (Vercel)	React / Next.js frontend	Serves the user interface pages to the browser over HTTPS.
Application Server (Local GPU server)	FastAPI backend running on Uvicorn, PyTorch runtime, APEX-Net model weights	Exposes the REST API, validates tokens, pre-processes images, runs inference, and generates heatmaps.

Node	Software Deployed	Responsibility
Supabase Cloud	Authentication and email service, PostgreSQL database, storage bucket	Manages accounts and sessions, stores profiles and prediction records, and stores X-ray and heatmap images.

The browser communicates with the frontend and the backend only over HTTPS. The frontend calls the backend REST API with the user's session token in each request, and the backend verifies this token before doing any work. The trained model is loaded into memory once when the backend starts, which avoids reloading it on every request. Secrets such as database keys and the token-signing secret are supplied through environment variables and are never hard-coded.

9. Applied Methodology

A user brings in a chest X-ray image, and the system analyses it with a deep learning model, shows the probability of fourteen thoracic diseases, and explains the result with a heatmap. Every step of the methodology maps directly to an action that a user performs in the web interface, so no technical setup is required on the user's side.

9.1 Workflow

The high-level workflow of the implemented APEX-Net system is shown below.

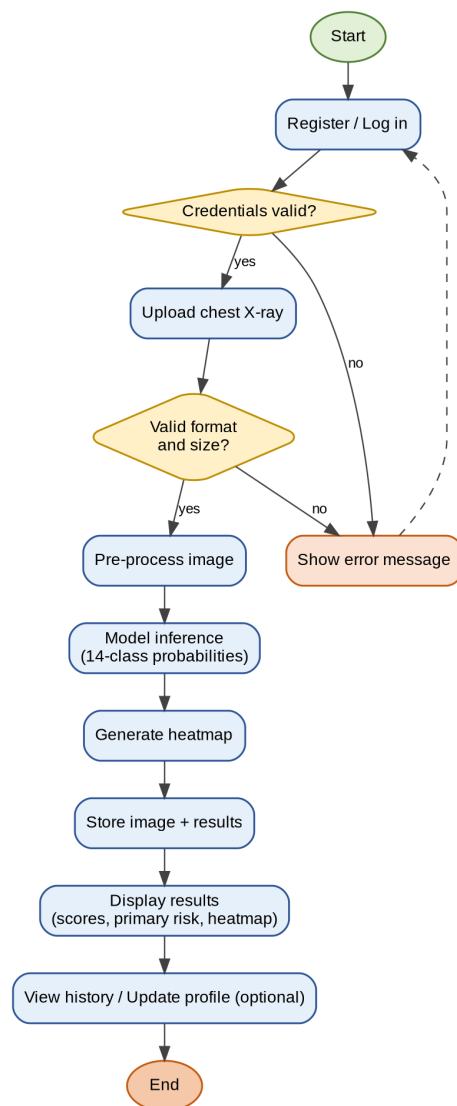


Figure-9 : High level workflow of APEX-Net

- The user registers an account or logs in with email and password.
- The user uploads a chest X-ray image (JPG, PNG, or DICOM).
- The backend validates the file and pre-processes the image.
- The APEX-Net model performs inference and produces 14 disease probabilities.
- The system identifies the primary risk and generates the heatmap overlay.
- The image, prediction, and heatmap are stored for the user.
- The results are displayed, and the user can revisit them later from the scan history.

9.2 Input

APEX-Net accepts input at several points in the workflow.

X-ray Image Input (for Analysis)

The central input of the system is a chest X-ray image provided by the user through the upload interface.

Table-4: Image input

Property	Accepted Value	Handling
File type	JPG / JPEG, PNG, DICOM (.dcm)	Any other file type (for example .txt or .pdf) is rejected immediately with the message “Unsupported file format”.
Number of images	One image per analysis request	Multiple uploads can be made one after another and are processed without blocking each other.
File size	Limited to a maximum size	Files larger than the limit are rejected before processing.
Content	Frontal chest radiograph	Unreadable or corrupted files are rejected with an error message.

Account and Profile Input

Table-5: Account and profile input

Input	Description
Full name	The name shown in the profile; stored in the Profiles table.
Email address	A valid email used to log in and to receive the confirmation email.
Password	A secret chosen by the user ; never stored as plain text.
Profile update	A new full name and/or a new password entered on the profile page.

Model Input

Internally, the model does not receive the raw file. It receives a pre-processed tensor produced from the validated image, as described in Section 9.3. For model training, the input is the labelled chest X-ray dataset (NIH ChestX-ray14) with 14 disease labels per image.

9.3 Processing

Image Validation and Pre-processing

1. The uploaded file is checked for an allowed type, a permitted size, and readability.
2. DICOM files are decoded to pixel data; JPG and PNG files are decoded as images.
3. The image is converted to the colour format expected by the network and resized to the fixed input resolution of the model (e.g. 224×224).
4. Pixel values are scaled and normalized using the same statistics that were used during training.
5. The image is converted into a tensor and handed to the Inference Engine. The original file is saved to object storage.

Model: DenseNet121 with Anatomical Prior Attention Module

The classification model uses DenseNet121 as its backbone. DenseNet connects each layer to all previous layers, which encourages feature reuse and keeps gradient flow healthy, and has shown strong performance on chest X-ray classification. The Anatomical Prior Attention Module (APAM) is inserted into the network to guide it towards

anatomically meaningful regions. It uses anatomical prior information about the chest, such as the lung and heart areas, to build a spatial attention map that re-weights the backbone's feature maps, so that features from relevant regions are emphasised and irrelevant background is suppressed. This mimics how a radiologist concentrates on specific anatomical regions when reading an X-ray.

The final layer produces 14 independent scores, one for each disease. A sigmoid activation converts each score into a probability between 0 and 1. Independent probabilities are used instead of a softmax because a single X-ray can show several diseases at the same time (multi-label classification).

Model Training Approach

1. The model is trained on the NIH ChestX-ray14 dataset, which contains labelled chest X-rays for the 14 thoracic conditions.
2. Training is balanced so that rare diseases are not overwhelmed by common ones, which addresses the imbalanced detection problem identified in Chapter 1.
3. After training, a decision threshold is tuned for each disease separately (per-disease sensitivity tuning), so that rare but critical conditions are not missed.
4. The best weights are saved and loaded by the backend at start-up. Training is a separate offline step and does not run inside the web application.

Inference and Interpretation

1. The pre-processed tensor is passed through the model and 14 probabilities are produced.
2. Each probability is compared with its disease-specific threshold to decide whether the disease is reported as likely.
3. If one disease probability is significantly higher than the rest, it is flagged as the primary risk.
4. A confidence score for the analysis is calculated and stored together with the full probability set.

Heatmap Generation

1. For the selected disease, the system computes the importance of each spatial location using the model's feature maps and gradients (Grad-CAM style class activation, combined with APAM attention).
2. The importance map is normalized and resized to the dimensions of the original image.

3. A colour map converts the values into a heatmap, where warmer colours mean stronger influence on the prediction.
4. The heatmap is blended over the original X-ray and saved as an overlay image.

Asynchronous Processing

Model inference can take noticeable time, especially when several users upload images together. The backend therefore processes analysis requests asynchronously: the request is accepted, the heavy computation runs without blocking other requests, and the interface shows a progress state until the result is ready. This keeps the application responsive and allows it to scale to multiple concurrent uploads.

Authentication and Session Handling

Every user action that involves personal data, such as uploading a scan or viewing history, requires a valid session token issued at login. The token is sent with each request, and the backend verifies it before processing. Requests with a missing, invalid, or expired token are rejected, and the user is redirected to the login page. Passwords are never stored in plain text; they are handled by the authentication service in hashed form.

Data Storage

All persistent data is stored in a PostgreSQL database, while image files are stored in object storage.

Table-6: Persistent data

Storage Item	What it stores
Profiles	Profile of each user: identifier, full name, and creation time.
Xray_Uploads	One record per uploaded image: owner, file name, file size, storage path, and upload time.
Predictions	One record per analysis: owner, linked upload, prediction data for all 14 diseases (JSONB), confidence score, and analysis time.
Object storage	The uploaded X-ray files and generated heatmap overlays, referenced through the storage path.

9.4 Output

Disease Prediction Results

The primary output of the system is the multi-label prediction for the uploaded X-ray. The result page shows:

- The probability score of each of the 14 thoracic diseases.
- The primary risk, when one disease probability is significantly higher than the others.
- The overall confidence score of the analysis and the time of analysis.

Heatmap Visualization

The system returns the X-ray with a heatmap overlay that highlights the regions that influenced the prediction. The user can compare the heatmap with the original image to understand where an abnormality may be located. This makes the result interpretable for general users, students, and clinicians.

Stored Scan Record

Each completed analysis is saved automatically. A saved scan stores the following:

Table-7: Contents of a saved scan

Field	Contents
File name	Name of the uploaded X-ray file.
Storage path	Location of the original image and the heatmap overlay in object storage.
Prediction data	The probabilities of all 14 diseases.
Confidence score	The overall confidence of the analysis.
Timestamps	Time of upload and time of analysis.

Scan History

The user can open the history page at any time to see all earlier scans with their timestamps, predictions, and heatmaps. This allows results to be tracked and compared over time. A user can also delete a scan, which removes both the record and its stored images.

Medical Disclaimer

Every result is accompanied by a notice that APEX-Net is a supportive screening tool and not a replacement for professional medical diagnosis, and that users should consult a qualified doctor for any health concern.

9.5 Code Implementation Overview

APEX-Net is a web-based platform that lets users upload chest X-rays and receive disease predictions with visual explanations. The system is built with a Python FastAPI backend that hosts the PyTorch model, a React / Next.js frontend, and Supabase for authentication, database, and storage. The following sections briefly describe how each major part is implemented. File and folder names below are indicative; exact names can differ from the names used in the original repository.

9.5.1 main.py (Backend Entry Point)

This file is the entry point of the backend. It creates the FastAPI application, applies the CORS policy that allows the React frontend to communicate with the API, registers the route modules for authentication, analysis, history, and profile under the /api prefix, and loads the trained model into memory at start-up so that every request can reuse it.

9.5.2 core/security.py (Security and Authentication)

This module protects the restricted routes. It provides a dependency that reads the session token from the request header, verifies it with the authentication service, and attaches the authenticated user to the request. Requests with a missing, invalid, or expired token immediately receive an HTTP 401 response. Registration, login, and confirmation emails are handled by the authentication service.

9.5.3 models/apam.py (Model Definition)

This module defines the APEX-Net network. It builds the DenseNet121 backbone, inserts the Anatomical Prior Attention Module (APAM), replaces the classifier with a 14-output layer, and provides a function that loads the trained weights. The same definition is shared by the training script and the inference service so that the saved weights always match the architecture.

9.5.4 dacnet.py (Training Pipeline)

This script runs offline and trains the model on the NIH ChestX-ray14 dataset. It prepares the data loaders, handles class imbalance, computes per-disease AUC during validation, saves the best checkpoint, and is used to tune the per-disease thresholds. It is not part of the running web application.

9.5.5 services/preprocess.py (Upload Validation and Pre-processing)

This service validates the uploaded file type, size, and readability, decodes JPG, PNG, and DICOM files, resizes and normalizes the image, and converts it to a tensor. It raises descriptive errors that the API converts into user-friendly messages such as “Unsupported file format”.

9.5.6 services/inference.py (Inference Engine)

This service runs the model on the pre-processed tensor without computing gradients for training. It applies the sigmoid function, compares each probability with its tuned threshold, determines the primary risk, and returns a structured result containing all 14 probabilities and the confidence score. Long-running work is executed asynchronously so that other requests are not blocked.

9.5.7 services/heatmap.py (Heatmap Generator)

This service computes the importance map for a disease from the model’s feature maps, normalizes and resizes it, converts it into a colour map, and blends it with the original X-ray. The resulting overlay image is saved to object storage and its path is returned to the caller.

9.5.8 routers/analysis.py and routers/history.py (Analysis and Scan History)

The analysis router receives the uploaded image, calls the pre-processing, inference, and heatmap services in sequence, saves the upload and prediction records, and returns the result. The history router lists a user’s previous scans, returns a single scan with its heatmap, and deletes scans. All database queries are filtered by the authenticated user’s identifier to guarantee per-user data isolation.

9.5.9 routers/profile.py (Profile Management)

This module returns the current user’s profile and allows the user to update the full name and change the password through a protected endpoint.

9.5.10 core/config.py (Configuration Management)

This module centralizes environment-specific settings such as the database and storage URLs, service keys, the token secret, allowed frontend origins, and the path of the model weights. Values are loaded from a .env file, so no credentials are hard-coded in the source code.

9.5.11 Frontend (React / Next.js)

The frontend is organized into pages for login, registration, upload and analysis, result viewing, scan history, and profile. Reusable components render the upload form, the probability table, and the heatmap viewer. A small API module sends requests to the backend with the session token and displays loading and error states, and the interface is responsive so that it works on desktop and mobile browsers.

10. User Manual and User Interface

10.1 Introduction

APEX-Net is a web-based assistant for screening chest X-ray images. It brings together three core capabilities in one place:

- **AI-Based Diagnosis:** upload a chest X-ray and receive the probability of 14 thoracic diseases.
- **Visual Explanation:** see a heatmap that highlights the regions of the X-ray that influenced the result.
- **Scan History:** keep a record of all your previous scans and results for later review.

This manual walks through every page and feature of the application so that any user, regardless of technical or medical background, can get started quickly.

10.2 System Requirements

- A modern web browser (Google Chrome, Firefox, Microsoft Edge, or Safari).
- An internet connection.
- A valid email address to create an account.
- A chest X-ray image in JPG, PNG, or DICOM format.

No installation is required on the user's computer.

10.3 Accessing APEX-Net

Open the browser and navigate to this address <https://apexnet2026.vercel.app/>. The first page shown is the Login page.

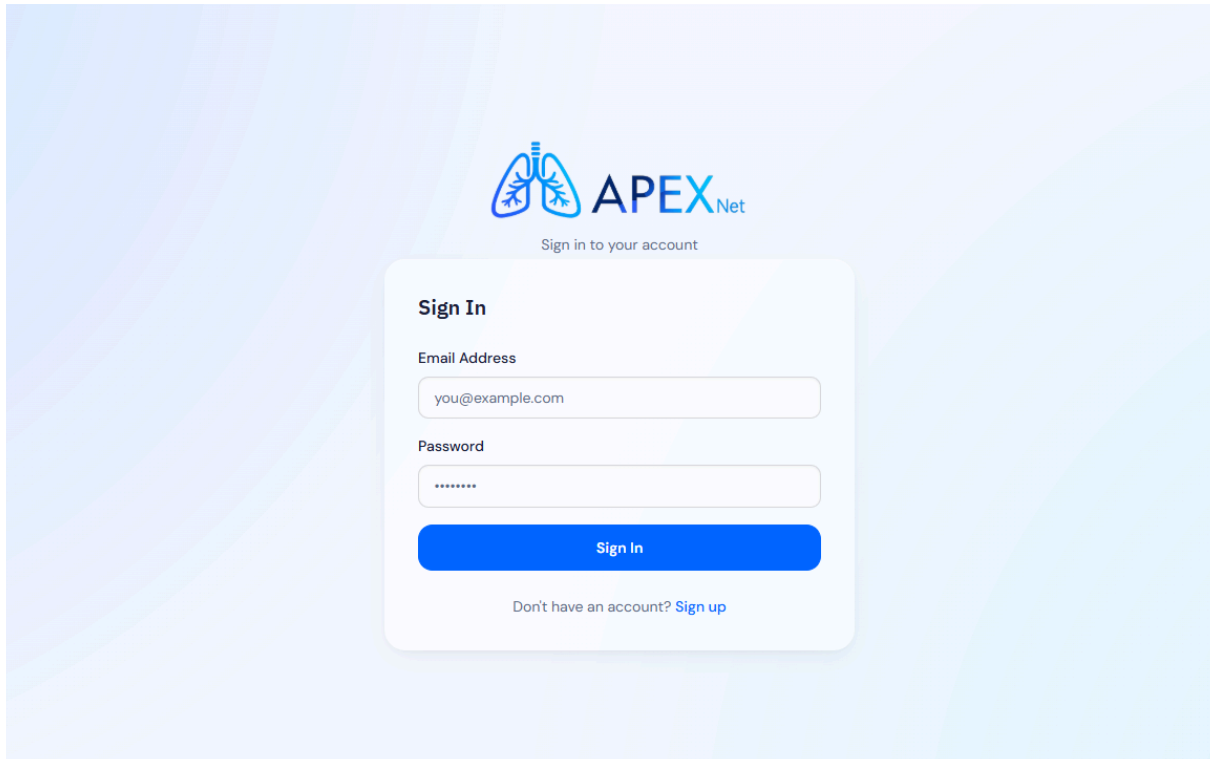


Figure-10 : Login Page

10.3.1 Source code repository links

Backend and model config: <https://github.com/habi8/APEX-Net>

Frontend : <https://github.com/habi8/apex-net-frontend>

10.4 Registering a New Account

If you do not have an account, you can create one from the Login page.

Steps:

- On the Login page, click “Sign up” (or “Register”).
- Fill in the following fields: **Full name**, **Email address**, and **Password**.
- Click **Register**.
- A message confirms that the account was created. Check your email inbox and open the confirmation email to activate the account.

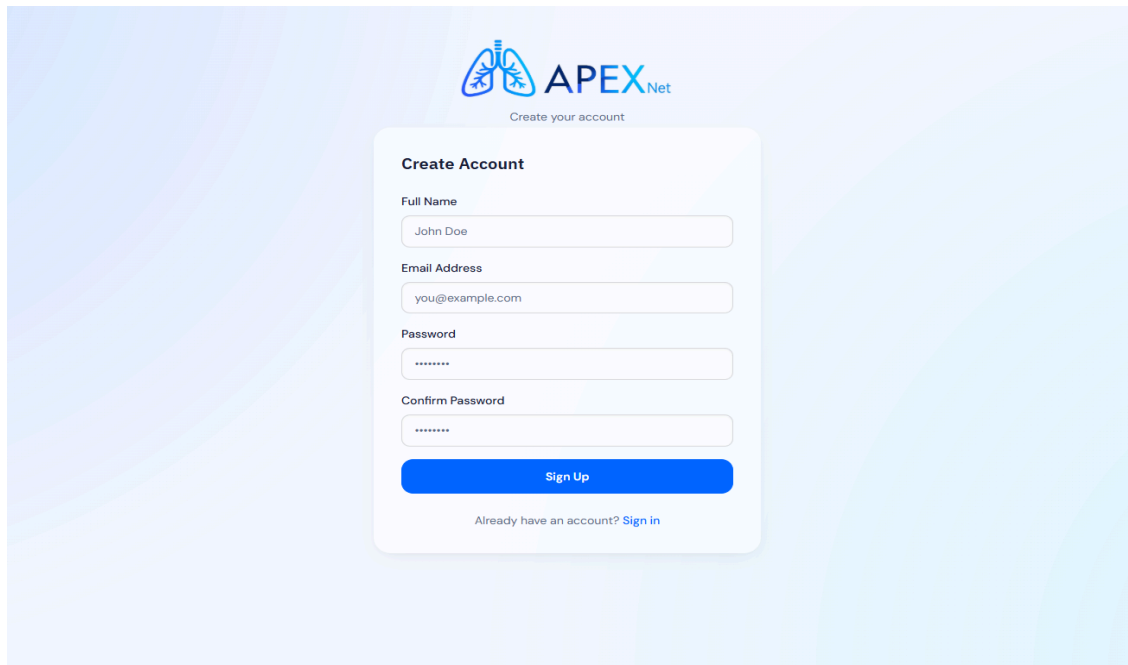


Figure-11 : Registration form

Note: if the confirmation email is not visible in the inbox, check the spam folder.

10.5 Logging In

Steps:

- On the Login page, enter your **Email** and **Password**.
- Click **Login**.
- If the credentials are correct, you are taken to the main page of the application.

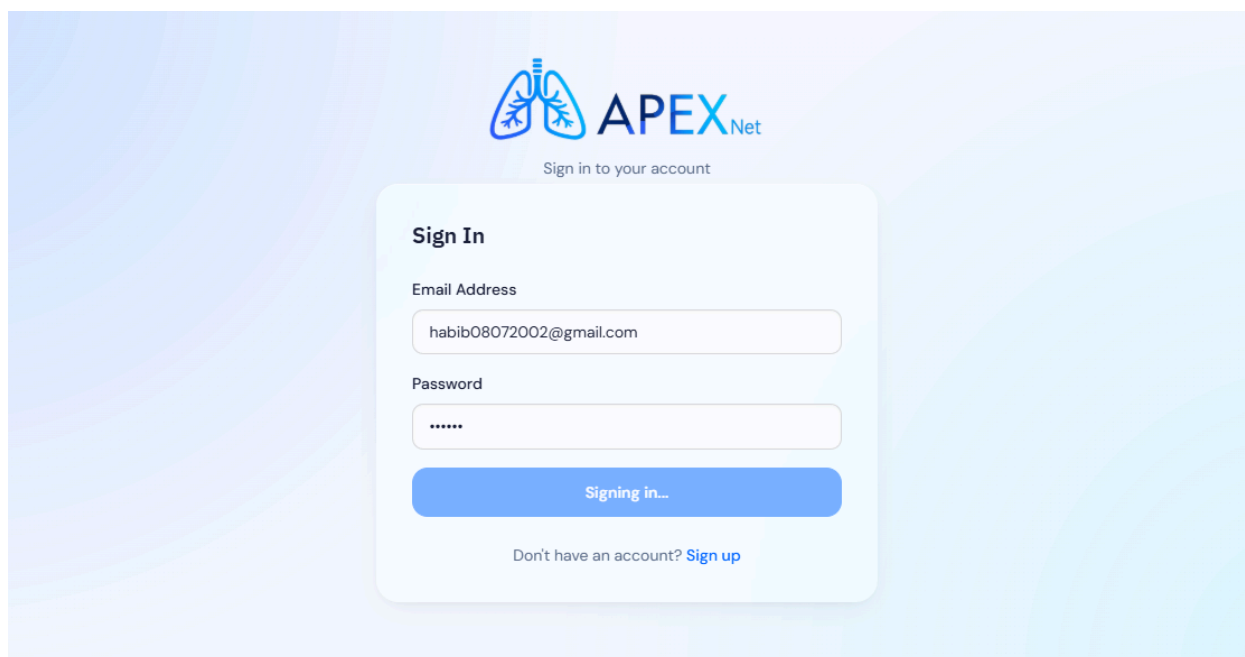


Figure-12 : Login page with credentials entered

If login fails, one of the following messages is shown:

Table-8: Login messages

Message	Cause
Invalid credentials	The email or password is wrong.
Email not confirmed	The account is registered but the confirmation link has not been opened yet.
Session expired	Your session ended; log in again.

10.6 The Application Layout

After logging in, the main interface has three areas:

- **Header (top bar):** shows the APEX-Net name and logo, your name, and a Logout button.
- **Navigation bar:** a row of links for switching between the pages of the application.
- **Main content area:** the active page is displayed here.

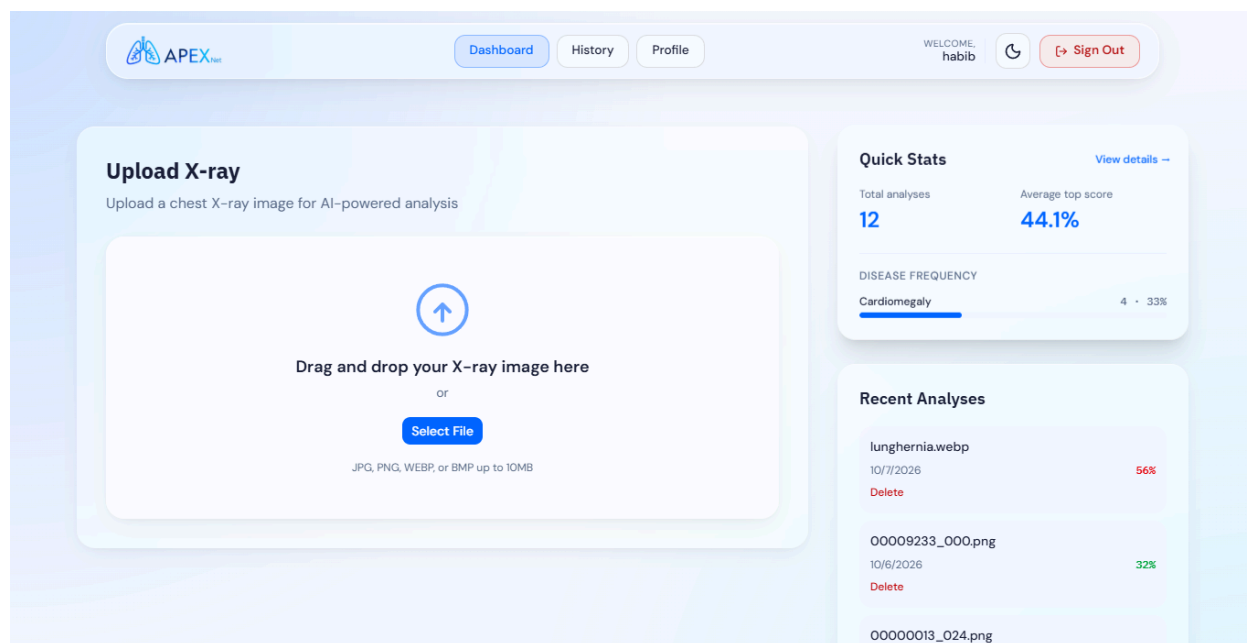


Figure-13 : Application window after login

Table-9: Navigation items

Item	What it does
Analyze	Upload a chest X-ray and get predictions with a heatmap.
Quick Stats	See stats based on previous results
History	View, open, and delete your previous scans.
Profile	View and update your name and password.
Logout	End your session securely.

10.7 Uploading and Analyzing an X-ray

Steps:

1. Open the **Dashboard** page.
2. Click **Choose file** (or drag and drop your image into the upload area).
3. Select a chest X-ray in JPG, PNG, or DICOM format. A preview of the image is shown.

4. Click **Analyze**. A progress indicator is displayed while the system processes the image.
5. When processing finishes, the results show automatically.

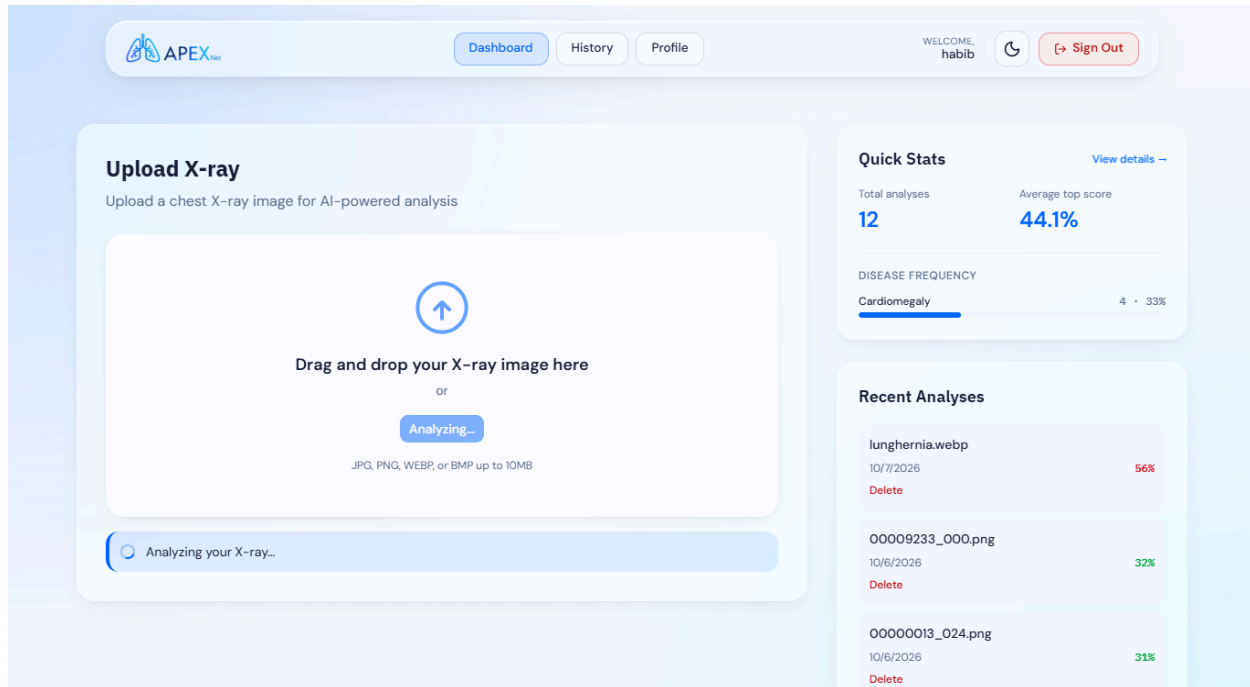


Figure-14 : Upload page with a selected X-ray

If the file type is not supported, the message “Unsupported file format” is displayed and nothing is processed. Choose a valid file and try again.

10.8 Understanding the Results

10.8.1 Disease Probabilities

The result page lists the 14 thoracic conditions: Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Emphysema, Fibrosis, Hernia, Infiltration, Mass, Nodule, Pleural Thickening, Pneumonia, and Pneumothorax. Each condition is shown with its probability. A higher probability means the model considers the condition more likely.

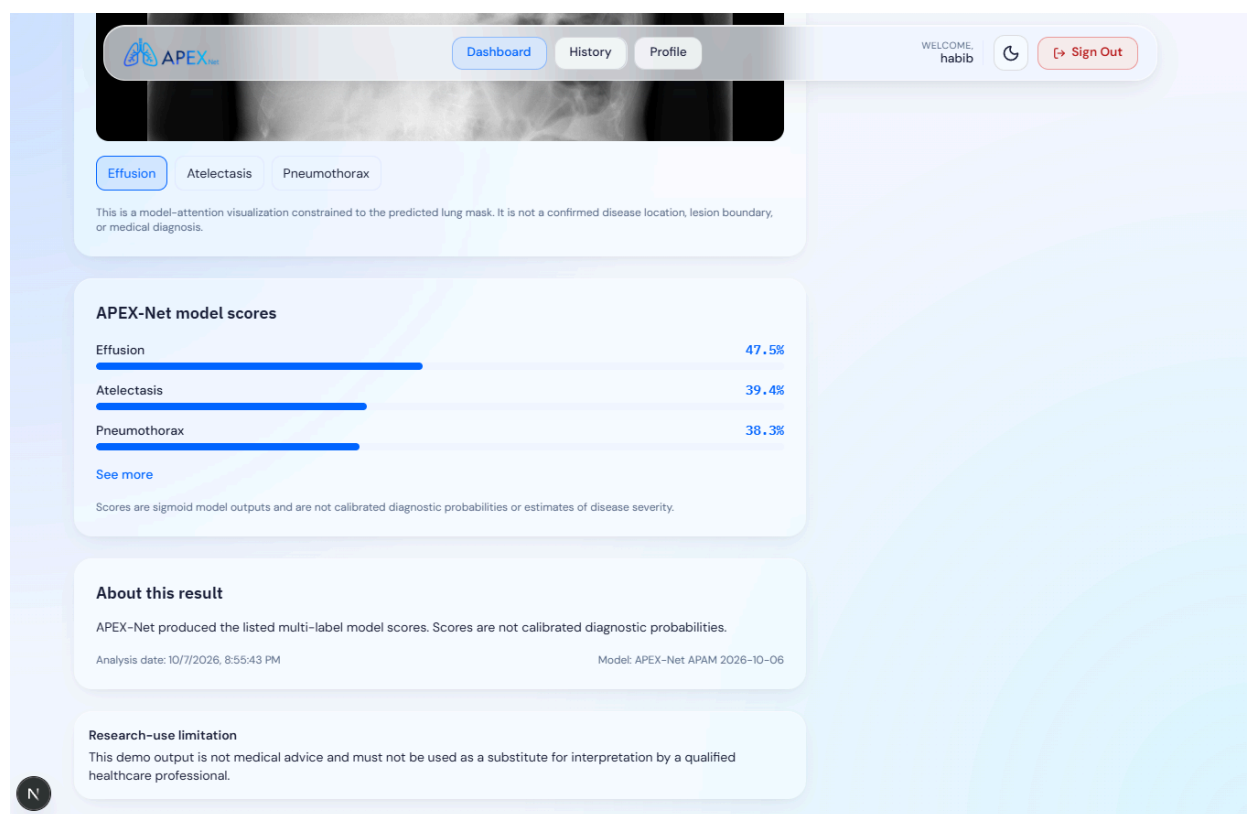


Figure-15 : Prediction result with disease probabilities

10.8.2 Primary Risk

When one disease has a probability that is significantly higher than the others, it is highlighted as the primary risk. This is meant to draw attention to the most important finding; it is not a final diagnosis.

10.8.3 Heatmap

Next to the probabilities, the original X-ray is shown with a coloured heatmap overlay. Warmer colours (red and yellow) mark regions that influenced the prediction the most, and cooler colours (blue) mark regions with little influence. Use the heatmap to see where the model found signs of concern.



Figure-16 : Heatmap overlay on the chest X-ray

Important: APEX-Net is a screening support tool and not a replacement for a doctor. Always consult a qualified medical professional for diagnosis and treatment.

10.9 Scan History

The **Scan History** page lists all your previous analyses with the upload time, file name, and main prediction.

Steps:

1. Open **Scan History** from the navigation bar.
2. Click a scan in the list to open its full result, including the probabilities and the heatmap.
3. To remove a scan, click **Delete** and confirm. The scan and its images are removed permanently.

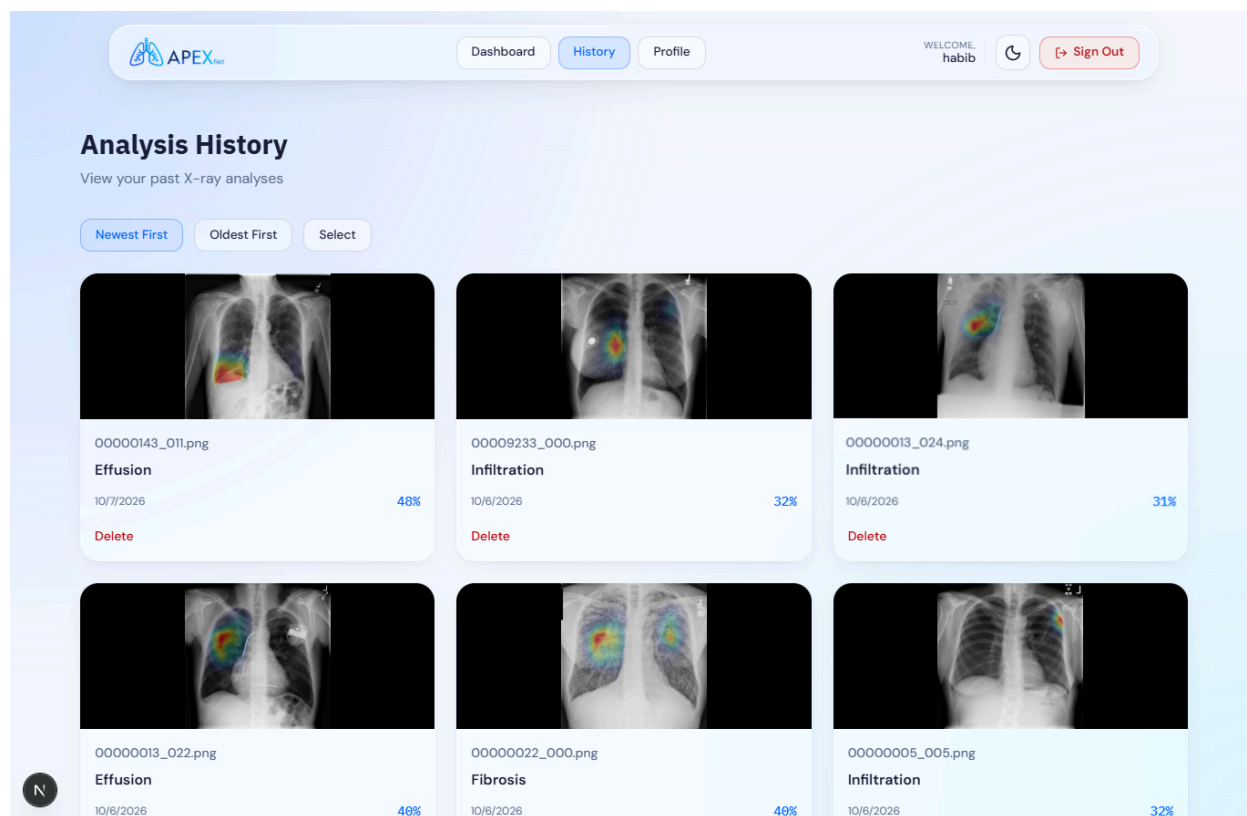


Figure-17 : Scan History page

10.10 Updating Your Profile

Steps:

1. Open the **Profile** page.
2. To change your name, edit the **Full name** field and click **Save**.
3. To delete an account, click **Delete Account** under the Danger Zone area. Accounts once deleted cannot be recovered.
4. A message confirms that the changes were saved.

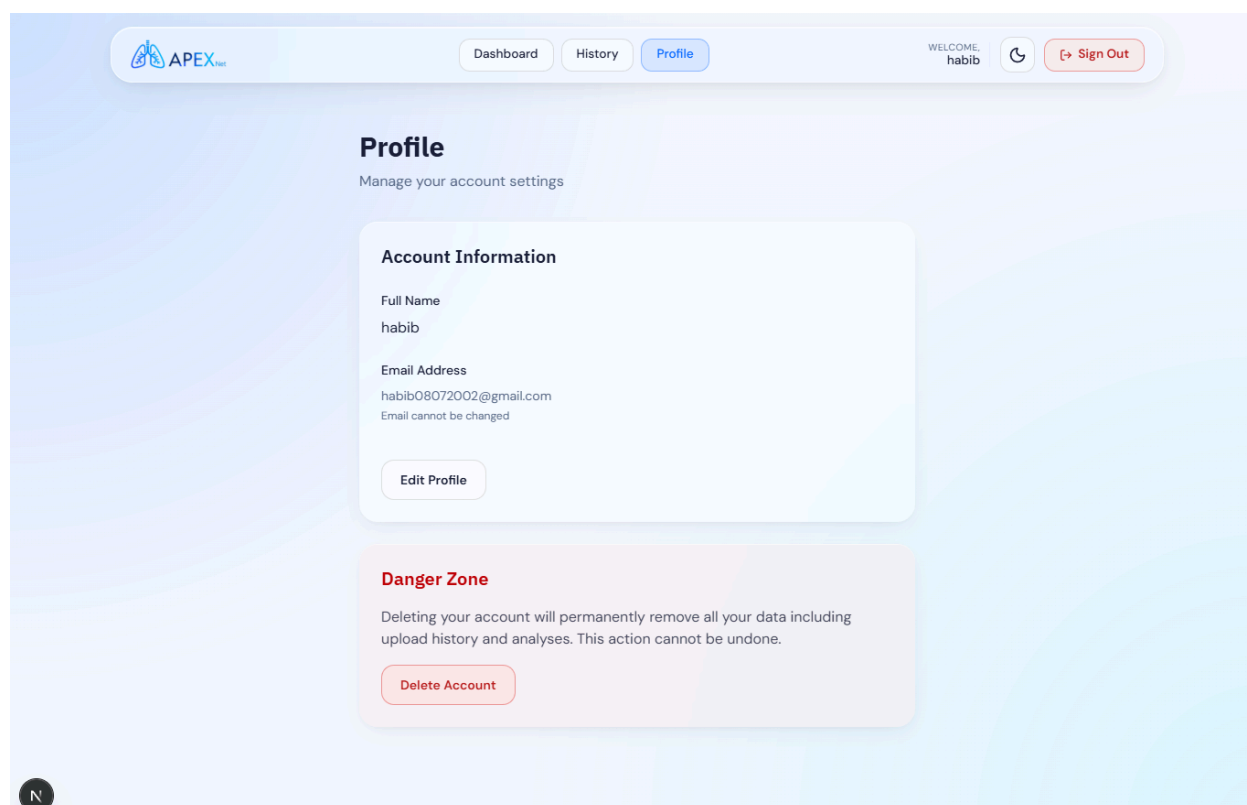


Figure-18 : Profile page

10.11 Logging Out

Click **Logout** in the header to end your session. For your privacy, always log out when using a shared computer.

10.12 Common Messages and Troubleshooting

Table-10: Common messages

Message / Problem	What to do
Unsupported file format	Upload a JPG, PNG, or DICOM chest X-ray.
File too large	Upload a smaller image file.
Analysis is taking long	Wait for the progress indicator; large images or many users at once can slow processing.
Session expired	Log in again and repeat the action.

Message / Problem	What to do
Heatmap not shown	Reopen the scan from Scan History; if it still does not appear, upload the image again.

11. Preliminary Test Plan

In this chapter, a high-level description of testing goals and a summary of items and features to be tested for APEX-Net are presented.

11.1 High-Level Description of Testing Goals

Serving as a bridge between users and the development team, software testing ensures the system is reliable, accurate, and user-friendly. For APEX-Net, the primary goal is to validate the correctness of AI predictions, system stability, and usability of the web application.

The goals of high-level software testing are given below:

- Ensure valid chest X-ray images are uploaded and processed correctly.
- Verify AI model generates accurate multi-label disease predictions.
- Confirm heatmap visualization correctly highlights relevant regions.
- Ensure invalid inputs are handled gracefully.
- Evaluate system performance under multiple concurrent uploads.
- Validate responsiveness and usability of the web interface.

11.2 Item Pass/Fail Criteria

If the system's output matches the expected result, the test case is considered successful. Otherwise, it is marked as failed.

Test Cases:

The following table provides the test cases that need to be executed for client-side and system testing:

Table-11: Testing

Test ID	Test Cases	Input Test Data	Steps to be Executed	Expected Result
T1	Valid X-ray Upload	Chest X-ray (.jpg/jpeg/.png/.dcm)	i) Upload valid image ii) Confirm upload	Image accepted and ready for analysis
T2	Invalid Input Handling	Unsupported file (.txt/.pdf)	i) Upload invalid file	Error message: "Unsupported file format"
T3	AI Prediction Output	Valid chest X-ray	i) Upload image ii) Run analysis	Multi-label disease predictions displayed
T4	Heatmap Visualization	Valid chest X-ray	i) Upload image ii) Run analysis	Heatmap displayed highlighting affected regions
T5	UI Workflow	Multiple uploads	i) Upload image ii) View result iii) Navigate history	Smooth workflow; results displayed correctly
T6	Performance Under Load	Batch of X-ray images	i) Upload multiple images simultaneously	System processes all without crash; acceptable response time
T7	Prediction Consistency	Same image (multiple runs)	i) Upload same image multiple times	Similar predictions generated with minimal variation
T8	Data Storage Verification	Valid upload + prediction	i) Upload image ii) Run analysis	Data stored correctly in database (image + prediction results)

This preliminary test plan ensures that APEX-Net is tested for functionality, reliability, performance, and usability before final deployment.

12. Conclusion

This report presents the design and technical foundation of **APEX-Net**, an AI-powered web application for thoracic disease detection using chest X-rays. The system's objectives, requirements, and workflows have been clearly defined, along with its architecture and data structure.

APEX-Net integrates deep learning with a user-friendly interface to provide multi-label disease predictions along with visual heatmaps, improving interpretability and usability. Its lightweight, web-based design makes it suitable for deployment in low-resource settings, where access to expert radiologists is limited.

Overall, APEX-Net serves as an effective tool for preliminary screening and supports better understanding of medical imaging through explainable AI.

Future Development Scope:

- Integration of more advanced deep learning models to improve diagnostic accuracy
- Integration of an image classification model for precise input validation
- Support for additional medical imaging modalities such as CT scans
- Real-time optimisation for faster inference and response
- Deployment as a mobile or cross-platform application
- Incorporation of clinician feedback for continuous model improvement
- Enhanced explainability techniques beyond heatmaps
- Integration with hospital or clinical information systems

In conclusion, APEX-Net demonstrates a practical and scalable approach to making AI-assisted medical diagnosis more accessible, interpretable, and reliable.

13. References

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[The End]